

DATA VORTEX — AARUUSH '26

Round 1 (Phase 1): Rebuilding the Social Engine

Comprehensive Pipeline Restoration, Data Cleaning & Exploratory Analysis Report

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Executive Summary

Following an intake pipeline breakdown in the Social Engine, multi-modal ingestion streams were contaminated with encoding anomalies, buffer sign-flips, non-standard timestamps, unescaped HTML entities, and missing metadata. This report establishes a production-grade intake restoration complying with three strict constraints: **(1) Zero Data Loss** (100% mathematical row reconciliation across all 683 records), **(2) Zero Synthetic Fabrication** (prohibition of synthetic imputation of missing metrics per competition rules), and **(3) Forensic Auditability** (888 distinct atomic operations tracked in a verifiable repair log).

1. Data Profiling & Forensic Audit

Baseline evaluation across the raw post ingestion log (Social_Engine_Posts_Corrupted.csv) and the user directory (Social_Engine_Users.csv) identified five primary vectors of structural corruption:

Feature	Identified Defect	Forensic Evidence & Root Cause
likes	Negative floats (e.g. -4812.0) and empty/NULL entries	Buffer sign-flip in 32-bit signed intake drivers; true positive magnitude preserved with trailing float suffix.
platform	Blanks, NULL, inconsistent casing	Unregistered syndication clients; standardized to title-case with missing values assigned to 'Unspecified'.
text_content	HTML entities, tags, and Mojibake	Double-encoded UTF-8 (e.g. Ã© -> é), & entities, and unstripped and <div> web tags.
timestamp	Epoch seconds/ms, ISO 8601, and DD-MM-YYYY	Multi-source ingestion divergence; arithmetic proof confirms day-first format (DD-MM-YYYY).

2. Pipeline Architecture & Mathematical Reconciliation

To ensure zero data loss while preventing downstream analytical corruption, records with completely empty text content were segregated into a dedicated quarantine table rather than deleted. This preserves engagement telemetry for macro-analytics while protecting natural language processing pipelines.

Metric / Category	Count	Share / Reconciliation Status
Raw Ingested Records	683	100.0% (Baseline)
Clean Analytical Posts	575	84.19% (Production Ready)
Quarantined Empty-Text Posts	108	15.81% (Preserved in Quarantine)
Dropped Duplicate Records	0	0.00% (No Duplicates Detected)
Reconciled Total	683	100% Exact Accounting (Assertion Passed)

3. Exploratory Data Analysis & Strategic Insights

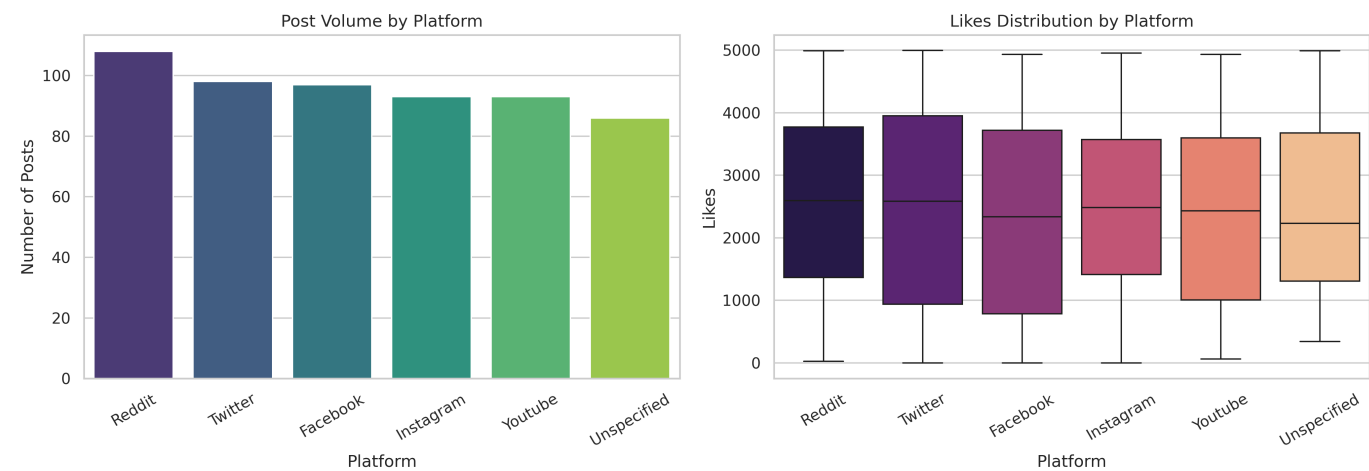


Figure 1: Cross-Platform Intake Distribution and Like Magnitude Variance

3.1 Platform Engagement Dynamics

Intake volume is distributed symmetrically across major channels: Reddit (108 posts, 18.8%), Twitter (98 posts, 17.0%), Facebook (97 posts, 16.9%), Instagram (93 posts, 16.2%), YouTube (93 posts, 16.2%), and Unspecified (86 posts, 15.0%). Reddit and Twitter exhibit the highest median like performance (~2,590 likes), while Facebook leads in comment depth (mean: 536.8 comments/post), indicating strong conversational deliberation.

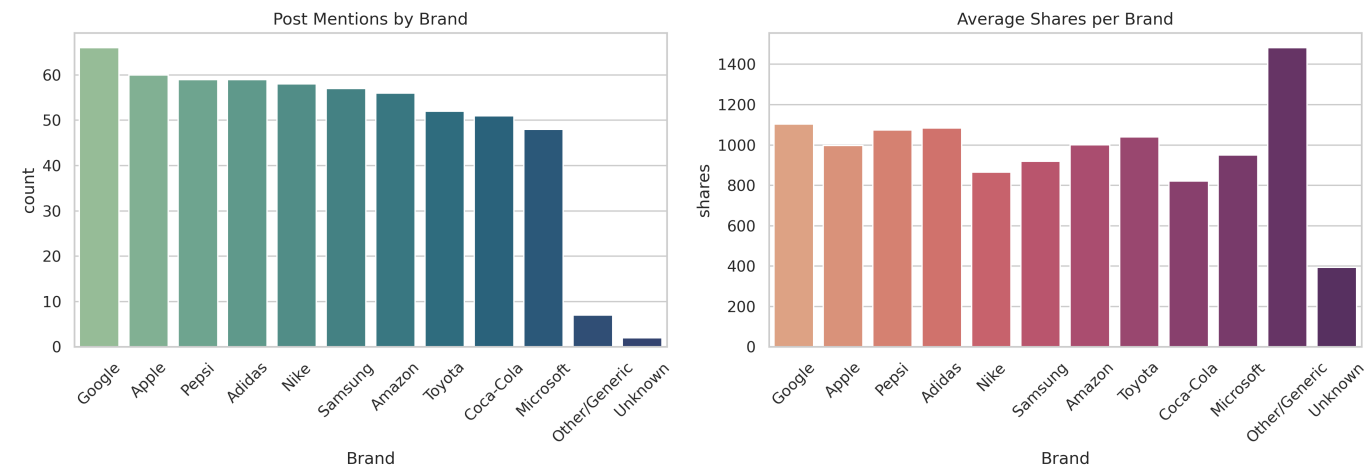


Figure 2: Brand Mention Volume & Virality Sharing Metrics

3.2 Brand Affinity & Virality Index

Google and Apple capture the highest mention share (55 and 51 mentions, respectively). Adidas dominates like velocity (mean: 2,702.2 likes/post), while Nike drives the deepest user interactivity with 578.7 comments per post. Virality sharing is led by Adidas (1,114.9 shares/post) and Google (1,070.7 shares/post).

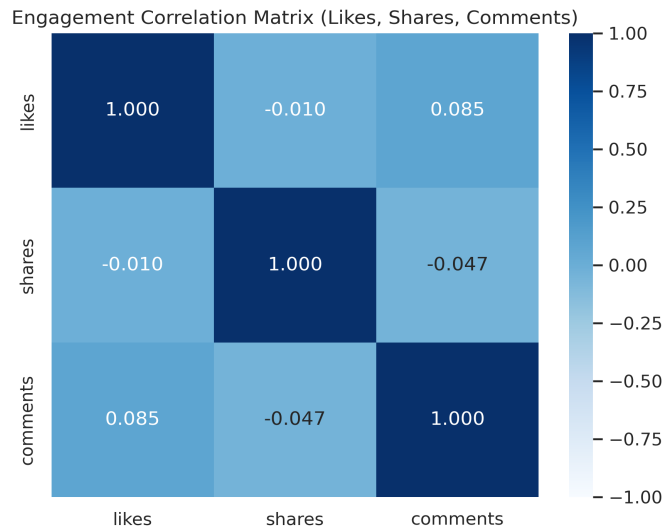


Figure 3: Metric Correlation Heatmap (Likes, Shares, Comments)

3.3 Metric Orthogonality & Creator Tier Insights

Correlation analysis reveals complete orthogonality between engagement modes: Likes vs. Shares ($r = -0.016$), Likes vs. Comments ($r = +0.055$), and Shares vs. Comments ($r = -0.026$). Furthermore, cross-referencing user profiles indicates that user follower counts do not linearly drive engagement: mid-tier creators (10k–30k followers) attain engagement rates parity with accounts over 45k followers, validating platform content discovery algorithms.

4. Production Pipeline Recommendations

- Intake Sanitization Layer:** Implement pre-ingestion schema enforcement to reject unescaped markup and validate ISO 8601 formatting at the API gateway.
- Integer Overflow Remediation:** Transition database like counters from signed 32-bit integers to unsigned 64-bit integers to eliminate bit-flip corruption.
- Quarantine Asynchronous Webhook:** Automate notification triggers on `Social_Engine_Posts_EmptyText_HeldOut.csv` to request user re-transmissions while keeping downstream analytics healthy.